DAY 8 OF 30

Bronze, Silver, Gold

The medallion architecture.
What each layer is really for.

Raw **Cleaned** **Business-ready**

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THE PROBLEM

One big "clean and report" job breaks badly.

1	**Source changes**	Everything fails at once
2	**A bug in logic**	Raw data is already gone
3	**New question**	Start again from the source
4	**An audit**	No record of what arrived

When raw, clean and reporting data are mixed, **every fix is a rebuild**.

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THE SIMPLE VERSION

Raw vegetables, chopped, cooked.

A kitchen does not cook straight from the delivery crate. It stores, washes and chops first, then cooks to order.

Data works the same way, in **three stages you can inspect separately**.

✓ **Bronze**	Exactly as it arrived
✓ **Silver**	Cleaned and consistent
✓ **Gold**	Shaped for a business question

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HOW IT WORKS

What happens in each layer

1. **Bronze**: land the data as-is and keep its history
2. **Silver**: fix types, remove duplicates, apply rules
3. **Silver**: join sources into shared, trusted tables
4. **Gold**: aggregate and model for reports and KPIs

The point

When logic changes, you rebuild from Bronze. You do not go back and ask the source again.

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FOR ENGINEERS

Decisions that shape the build

Layers are a contract, not a rule

Three is a convention. What matters is that each layer has a clear promise about quality.

Where layers live

Microsoft's guidance is one lakehouse per layer, each in its own workspace, so access and ownership stay clean.

Gold can be a warehouse

If consumers think in SQL, serve Gold from a Warehouse and keep Bronze and Silver in a Lakehouse.

Materialized lake views

Declare Silver and Gold as Spark SQL. Fabric stores them as Delta, orders the refreshes and applies quality rules.

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THE HONEST PART

Do you always need three layers?

WORTH IT WHEN

Several sources feed the same reports

Logic changes often

You must show what arrived and when

OVERKILL WHEN

One small, clean source

A short-lived prototype

Nobody will maintain three layers

> **The catch:** medallion fails when Silver quietly becomes a second Bronze. If Silver has no quality rules, you have added a layer and gained nothing.

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FROM REAL WORK

Medallion on a real project

A people-analytics platform built on Fabric with PySpark.

1. **Bronze**: raw employee data landed by daily pipelines
2. **Silver**: cleaned and conformed across HR sources
3. **Gold**: tables shaped for organisation KPIs
4. **Semantic models** read Gold for reporting

Each layer had **one job**, which keeps changes contained.

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SAVE THIS

Medallion in five lines

1. **Bronze** keeps what arrived, untouched
2. **Silver** is where data becomes trustworthy
3. **Gold** answers business questions
4. **Rebuild from Bronze**, never from the source
5. **Each layer** needs a clear promise

Tomorrow: Data Factory pipelines. The scheduler that moves everything.

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